PathwayAI | Dutchess County

Lyme Disease Burden & Community Action

Prepared 2026-11-02 · Public-data planning brief · Version 1.9

For local planning discussions · Proposed actions for review

Reported cases · 2024	County rate per 100,000 · 2024	State rate per 100,000 · 2024
1,060	355.5	109.8

Illustrative economic burden estimate

USD 2.89 million

Optional 2025 purchasing-power comparison · 11 available CPI months · general inflation

Original study basis · 2016 dollars	Optional 2025 comparison · approximately 1.34×
USD 2.15 million	USD 2.89 million

Study-based illustration, not measured county spending or predicted savings. Sources [1, 5, 11]; calculation and methods: page 4.

Mid-Hudson counties and state comparison

Area	Reported cases · 2024	Rate per 100,000 · 2024
Dutchess	1,060	355.5
Orange	964	237
Putnam	294	299.2
Rockland	461	135.3
Sullivan	385	482.6
Ulster	841	461.2
Westchester	1,125	113.2
New York State	21,632	109.8

All comparison rates use the same NYSDOH 2024 published source and reporting year; reported rates are not personal risk estimates.

Healthcare resources and population context

Indicator	Value	Period
Primary-care physicians	233	2023
Hospitals	2	2023
Hospital beds	404	2023
Uninsured residents under age 65	4.7%	2022
Poverty (population with poverty status determined)	8%	2023
Language other than English at home · age 5+	16.2%	ACS 2020–2024

General resource/population context, not Lyme-attributable need, appointment availability or interpreter demand. Selected insurance/poverty fields supply no margin of error.

Reported Lyme rates by year · per 100,000 residents

2022	2023	2024
394.6	358.9	355.5

County of residence, not exposure location. Pre-2022 rates excluded because surveillance definitions changed. Different sources do not establish causes. References and limitations: page 4.

From evidence to community action

Published components · per participant · 2016 dollars

Medical: USD 1,333; other expenses and productivity combined: USD 699; total: USD 2,032.

The combined component is a derived residual, not a separately reported mean. A separate caregiving amount is not established here; do not add components to the total again.

Support priority: Getting to care

Question: Should a county care-navigation team investigate a small ride-navigation pilot with willing residents?

Possible response: Ask willing residents and transport providers to review ride eligibility, accessibility and appointment coordination.

Who could help: Patient and caregiver representatives, transport services and care navigators

Actions to discuss · Public health departments

- Share tick-prevention information through local events and trusted community channels.
- Review county surveillance alongside service availability; ask residents which barriers need attention.
- Bring residents and potential partners together to choose one support priority and agree how to follow up.

Four areas for community action

Prevention education and tick-bite awareness: Share established prevention information through trusted local channels.

Timely access to healthcare: Help residents find care and ask about appointments, costs, transport and interpreters.

Financial and caregiving burden: Recognize spending, missed work, unpaid care and support needs beyond medical bills.

Language access and community resources: Make information easier to understand and connect residents with existing local support.

Resident priorities to discuss

Confirm priorities with residents and caregivers before choosing a response. No county-specific patient-reported findings are established in this brief.

Before starting a pilot

- Confirm local demand with residents and caregivers, including assisted/offline participation.
- Verify current services, eligibility, interpreter access and appointment capacity.
- Agree who is responsible and estimate staff time, program costs and a funding source.
- Define consent, minimal data collection, access, retention and deletion before any collection.

Check whether people received support

Choose a common follow-up period, such as 30 days. Record the starting situation, help received and needs still unmet. Keep unanswered questions unknown.

For this priority: Requested and completed accessible rides; appointments attended; travel spending; support still needed

Further evaluation is needed to establish health benefits or savings. Actions are proposals for local review, not adopted programs or commitments.

Professional review

Public-health, health-economics and Lyme-care reviewers: please comment on source clarity, cost assumptions, practical usefulness and missing local factors.

Email feedback: pathwayai.feedback@gmail.com

Local planning & research context

What This Means for Local Planning

1. Prevention outreach

Dutchess's 2024 reported rate was 355.5 per 100,000, above the state rate of 109.8. Where could trusted local channels help share prevention information? [2]

2. Care access

4.7% of residents under age 65 were uninsured in 2022. Could care navigators help residents ask about sliding fees and coverage options? This is general population context, not Lyme-specific need or available appointments. [4]

3. A local action to review

Should a county care-navigation team investigate a small ride-navigation pilot with willing residents? Confirm needs with residents and check existing services before selecting a response.

Planning questions for local review; population indicators do not establish Lyme-specific need or available services.

What published research adds

External study populations provide context, not county patient findings.

Hook et al. (2022) [5]

901 patient-cost participants; 613 societal-cost participants · Recruitment 2014–2016; 2016 dollars

Medical, nonmedical and productivity costs inform the existing illustration.

Population and limits: Selected reported cases in four high-incidence states; excluded non-English speakers and nonacute disease.

Adrion et al. (2015) [12]

52,795 Lyme patients; 263,975 matched controls · Claims 2006–2010; 12-month follow-up

Documents healthcare spending and utilization after Lyme diagnosis.

Population and limits: Commercially insured ages 0–64; not uninsured patients or a full household-cost estimate.

Johnson et al. (2014) [13]

3,090 selected survey respondents · Survey 2013; persistent symptoms after treatment

Documents reported work disruption, expenses and quality-of-life impacts.

Population and limits: Self-selected, self-reported persistent-illness sample; not all Lyme cases or a county disability rate.

How to use this evidence

Use these studies to discuss spending, work disruption and daily-life impacts with local residents. Hook remains the sole study-cost basis for the county illustration. Study estimates are not added together or applied as local disability rates.

County-specific patient findings would require a defined local collection and review process. References and detailed methods follow on page 4.

References & methods

Original study basis

1,060 reported cases × USD 2,032 study mean = USD 2,153,920 (2016 dollars).

Hook et al. (published 2022): 613 participants in societal analysis; reported cases recruited in four high-incidence states during 2014–2016. Societal: medical, nonmedical and productivity costs; components are already included.

Applying this study mean to reported county cases is an assumption; the result is not measured local spending. No adjustment for underreporting or local case mix. No causal savings or disability probabilities are inferred.

Hook excluded non-English speakers and nonacute disease. The study does not represent every local household or continuing lifetime costs.

Optional inflation comparison

General CPI factor approximately 1.34; calculations use the full-precision factor retained in the methods CSV. Adjusted study mean approximately USD 2,726 per participant (rounded for display).

2016 uses 12 monthly observations; 2025 uses 11 available months. October is missing; no imputation. This is not a verified official full-year annual average or medical-care inflation adjustment. It changes purchasing power, not case mix, treatment patterns or local prices.

Source register

1. Reported cases, county of residence · 2024

2. Reported rates per 100,000 · 2024

3. CDC surveillance interpretation; historical county file in app · 2019–2022 county file; separate from NYSDOH 2024

4. Healthcare resources, insurance and poverty context · 2022 insurance fields; 2023 healthcare and poverty fields; AHRF 2024–2025 release

5. Hook et al. Economic Burden of Reported Lyme Disease in High-Incidence Areas, United States, 2014–2016. Emerging Infectious Diseases. 2022. · 2016 dollars; recruitment 2014–2016

6. Shared Mid-Hudson regional assessment (Dutchess and Orange), Figure 147, page 219 · 2022–2023; regional assessment published 2025

7. Language spoken at home, age 5+ · ACS 2020–2024

8. Tick Task Force · Existing directory; current status requires verification

9. Mobile Health Unit · Existing directory; current status requires verification

10. Community Health Improvement Plan · Existing directory; current status requires verification

11. BLS CPI-U via supplied FRED file · 2016 and 2025 available-month comparison

12. Adrion et al. Health Care Costs, Utilization and Patterns of Care following Lyme Disease. PLOS ONE. 2015;10:e0116767. · Claims 2006–2010; 12-month follow-up

13. Johnson et al. Severity of chronic Lyme disease compared to other chronic conditions: a quality of life survey. PeerJ. 2014;2:e322. · Survey 2013; persistent symptoms after treatment

Verification and interpretation

Existing source snapshots; newer official releases have not been verified for this brief. Official rates are reproduced; exact rate-denominator vintage is not independently verified. Pre-2022 rates are excluded because surveillance definitions changed.

Census language context is an existing QuickFacts snapshot, not a fresh pull or a measure of English proficiency. HRSA field years are not substituted with newer Census periods. Initiative links require current local verification; no listing does not mean no services exist.

Public evidence and proposed planning actions only. No session stories, personal records or synthetic patient cards are included.

Discussion brief, not a county demand assessment, benefits decision, funded service or adopted county plan. No county or organizational endorsement is implied.